# JASPERGOLD FORMAL VRF

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**githubUrl:** https://github.com/vasanthasricharan/JasperGold-Formal-VRF

## Abstract

This project focuses on applying formal verification techniques during RTL verification using Cadence JasperGold. The design under verification is an FSM-based secure arithmetic processing module implemented in Verilog HDL. Formal verification is used to mathematically prove correctness by exploring all reachable states and input combinations, unlike traditional simulation which validates only limited test scenarios. The project verifies FSM transitions, reset behavior, protocol compliance, security handling, deadlock-free operation, unreachable states, and overflow conditions.

## Problem Statement

Simulation-based verification using Universal Verification Methodology tests only a limited set of input scenarios, so some corner-case bugs or illegal states may remain undetected.

## Project Objective

The objective of this project is to demonstrate the advantages of formal verification over traditional simulation-based verification using Cadence JasperGold. The project aims to prove design correctness mathematically by exhaustively exploring all possible states and conditions within the RTL design.

## Why Formal Verification?

- Explores all reachable RTL states.
- Detects hidden corner-case bugs.
- Identifies unreachable and illegal FSM states.
- Verifies deadlock-free operation.
- Improves verification coverage beyond simulation.
- Provides mathematical proof of correctness.

### Design Overview

The design is an FSM-based secure arithmetic module implemented in Verilog HDL. It performs arithmetic and logical operations based on the opcode input and includes security and power control logic.

### Inputs and Outputs

| Inputs | Outputs |
|---|---|
| clk | grant |
| rst | result[4:0] |
| req | valid |
| data_a[3:0] | error_flag |
| data_b[3:0] | |
| opcode[1:0] | |

| | |
|---|---|
| secure_access | |
| power_enable | |

### FSM State Description

| State | Encoding | Description |
|---|---|---|
| IDLE | 2'b00 | Waits for request signal. |
| PROCESS | 2'b01 | Performs operation based on opcode. |
| DONE | 2'b10 | Operation completed successfully. |
| ERROR | 2'b11 | Indicates security or power failure. |

### FSM State Diagram

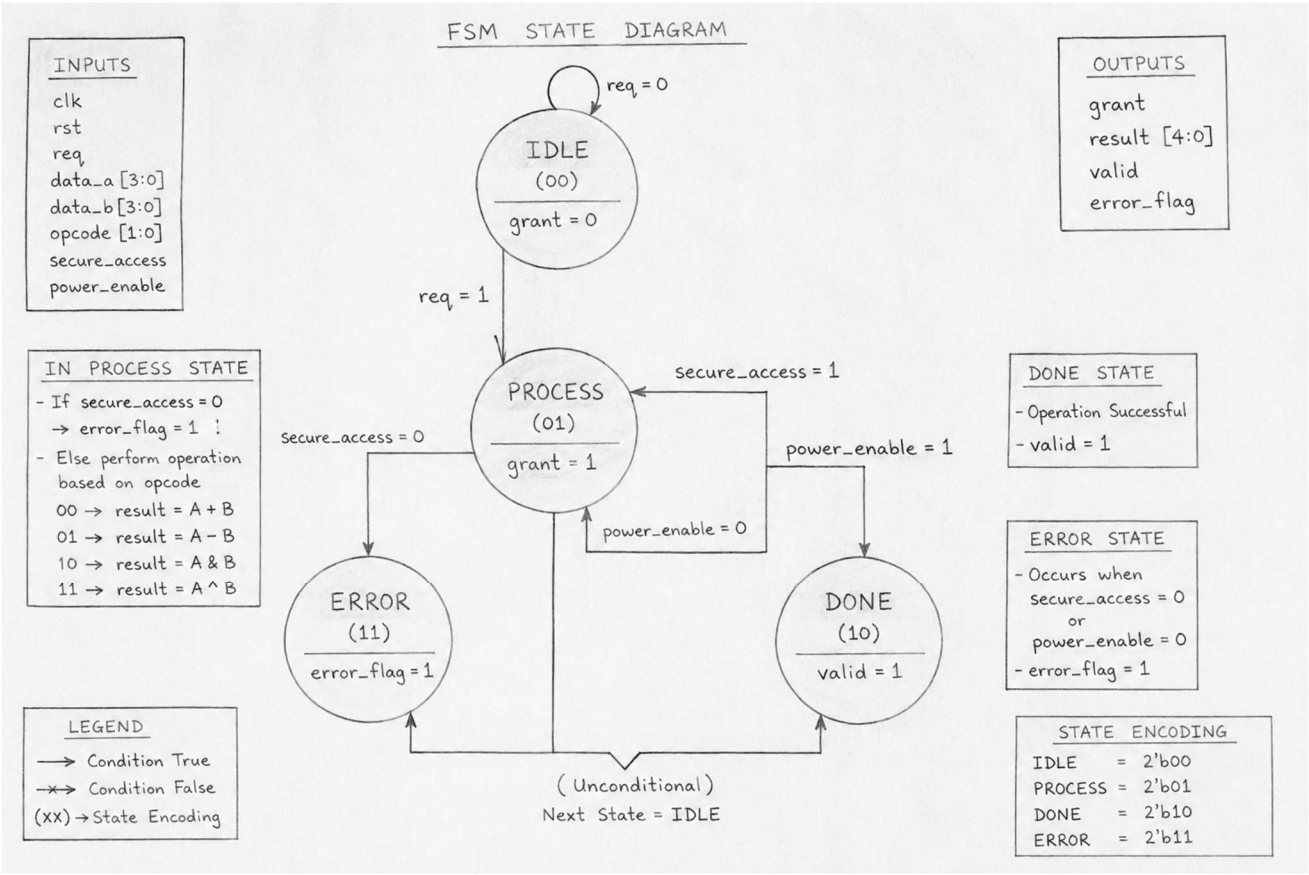



Figure: FSM State Diagram of Secure Arithmetic Module

### RTL Design Explanation

The RTL design consists of a finite state machine with four states: IDLE, PROCESS, DONE, and ERROR. The state register updates on the positive edge of the clock or reset. The next-state logic determines transitions based on request, power_enable, and security conditions.

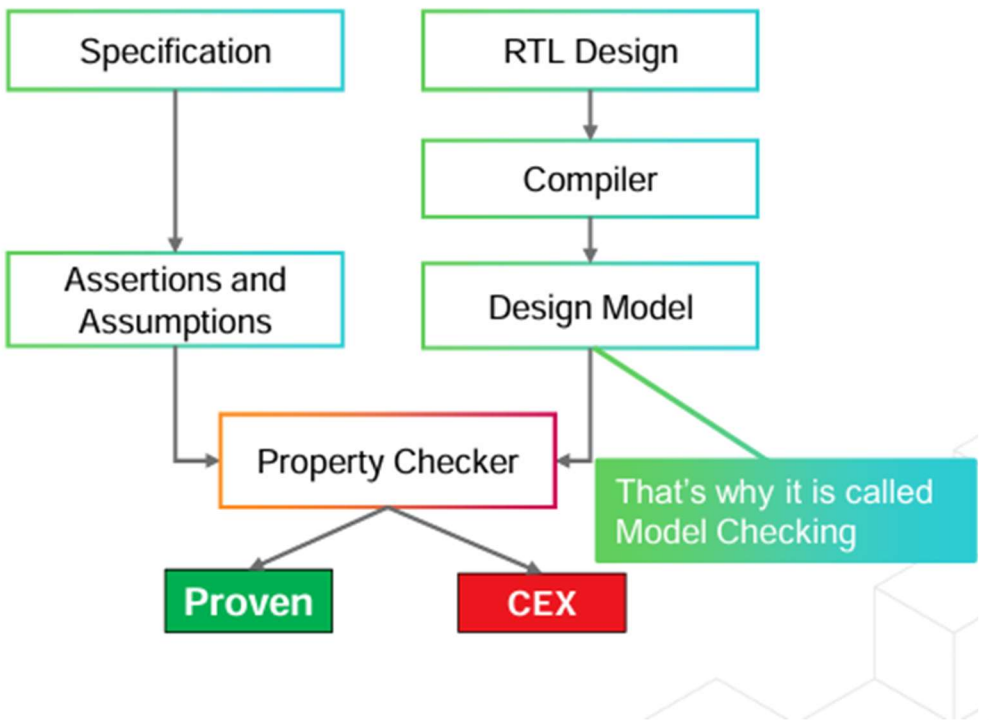
In the PROCESS state, arithmetic or logical operations are executed based on the opcode input: 00 performs addition, 01 performs subtraction, 10 performs bitwise AND, and 11 performs bitwise XOR.

## Verification Methodology

- RTL Simulation Verification
- Formal Verification using JasperGold
- FSM Transition Checking
- Reset Verification
- Security Verification
- Protocol Compliance Checking
- Deadlock Detection
- Unreachable State Analysis
- X-State Analysis

## Formal Verification Concept

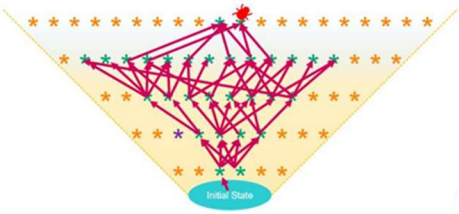
Formal verification explores all reachable design states to prove properties or find failures. Unlike simulation, which checks selected test vectors, formal verification mathematically analyzes every possible input combination and state transition.




## Simulation vs Formal Verification

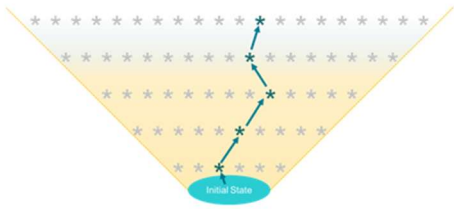
| Simulation Verification | Formal Verification |
|---|---|
| Checks limited test cases | Explores all possible states |
| Depends on testbench quality | Mathematically proves correctness |
| May miss corner-case bugs | Finds hidden and corner-case bugs |
| Sequential execution | Parallel symbolic exploration |
| Limited coverage | Exhaustive coverage |

**Formal**




Formal uses smart mathematical techniques to consider all relevant states.

**Simulation**




Simulation only covers some of the DUT states.

## Formal verification Report

```
CORNER CASE VERIFIED : MAX VALUE
NO DEADLOCK DETECTED

==========================================================================
|                          Jasper formal Vrf ANALYTICS REPORT
==========================================================================
| FUNCTIONAL VERIFICATION
|------------------------------------------------------------------------
| Total Operations Tested        : 64
| Passed Operations              : 40
| Failed Operations              : 24
| Functional Accuracy            : 62.50 %
|------------------------------------------------------------------------
| PROTOCOL VERIFICATION
| Successful Handshakes          : 32
| Failed Handshakes              : 32
| Protocol Efficiency            : 50.00 %
|------------------------------------------------------------------------
| FSM COVERAGE
| IDLE State Visits              : 32
| PROCESS State Visits           : 32
| DONE State Visits              : 0
| ERROR State Visits             : 0
| FSM Coverage                   : 0.00 %
|------------------------------------------------------------------------
| SECURITY VERIFICATION
| Unauthorized Access Attempts   : 64
| Blocked Access Attempts        : 32
| Security Integrity             : 50.00%
|------------------------------------------------------------------------
| X-PROPAGATION ANALYSIS
| X-State Injections             : 64
| X-State Detections             : 64
|------------------------------------------------------------------------
| CORNER CASE ANALYSIS
| Minimum Boundary Tests         : 4
| Maximum Boundary Tests         : 4
|------------------------------------------------------------------------
| DEADLOCK ANALYSIS
| Deadlock Checks                : 1
| Deadlock Failures              : 0
|------------------------------------------------------------------------
| OVERALL VERIFICATION CONFIDENCE : 40.62 %
==========================================================================

##########################################################################
#                          FINAL VERIFICATION CONCLUSION
##########################################################################
# RESULT : VERIFICATION NEEDS IMPROVEMENT
# JasperGold-Style Verification Analytics Completed Successfully
# Verification Methodology : Simulation + Assertion-Style Analysis
# Design Verification Flow Includes FSM, Protocol, Security, Power, and X-State Analysis
##########################################################################
```

## Simulation-based verification report:

```
==========================================================================
|                NORMAL RTL SIMULATION ANALYTICS REPORT                 |
==========================================================================
| FUNCTIONAL VERIFICATION
|------------------------------------------------------------------------
| Total Operations Tested      : 4
| Passed Operations            : 4
| Failed Operations            : 0
| Functional Accuracy          : 100.00 %
|------------------------------------------------------------------------
| PROTOCOL VERIFICATION
| Successful Handshakes        : 4
| Failed Handshakes            : 0
| Protocol Accuracy            : 100.00 %
|------------------------------------------------------------------------
| FSM STATE OBSERVATION
| IDLE State Visits            : 0
| PROCESS State Visits         : 4
| DONE State Visits            : 0
| ERROR State Visits           : 0
|------------------------------------------------------------------------
| SECURITY TESTING
| Security Checks              : 5
| Security Passes              : 4
| Security Failures            : 1
| Security Accuracy            : 80.00 %
|------------------------------------------------------------------------
| X-PROPAGATION ANALYSIS
| X-State Injections           : 1
| X-State Detections           : 1
|------------------------------------------------------------------------
| CORNER CASE ANALYSIS
| Minimum Boundary Tests       : 0
| Maximum Boundary Tests       : 0
|------------------------------------------------------------------------
| RESET VERIFICATION
| Reset Checks                 : 1
| Reset Passes                 : 1
| Reset Failures               : 0
| Reset Accuracy               : 100.00 %
|------------------------------------------------------------------------
| DEADLOCK ANALYSIS
| Deadlock Checks              : 1
| Deadlock Failures            : 1
|------------------------------------------------------------------------
| OVERALL SIMULATION CONFIDENCE  : 100.00 %
==========================================================================

##########################################################################
#                          NORMAL SIMULATION COMPLETED                   #
##########################################################################
# Verification Method : Directed RTL Simulation                          #
# Verification Scope  : Limited Stimulus-Based Testing                   #
# Verification Nature : Runtime Behavioral Observation                   #
# Coverage Type       : Testcase Dependent                               #
##########################################################################
```

## Comparative Analysis of Simulation and Formal Verification Reports

The normal RTL simulation report shows 100% functional and protocol accuracy because verification is performed only on a limited set of directed test cases. Although the simulation confirms correct behaviour for tested scenarios, several FSM states such as DONE and ERROR were never explored, and corner-case coverage remained limited. The report also indicates a deadlock failure despite successful functional execution, highlighting the limitations of stimulus-dependent verification.

In contrast, the JasperGold formal verification report performs exhaustive state-space exploration by analysing all possible input combinations and FSM transitions. Formal verification identified additional hidden behaviours including failed operations, protocol mismatches, unauthorized access conditions, and broader security-related scenarios that were not visible in normal simulation. The report also confirms that no deadlock condition exists under complete state exploration, demonstrating the strength of mathematical proof-based verification.

Overall, the comparison proves that simulation-based verification validates only expected runtime behaviour, while formal verification provides deeper analysis, improved state coverage, corner-case detection, and higher confidence in RTL design correctness.

## SystemVerilog Assertions (SVA)

SystemVerilog Assertions are used to verify critical properties such as correct FSM transitions, reset behavior, valid signal generation, and deadlock-free operation.

## Advantages of JasperGold

- Fast bug detection
- Exhaustive state-space exploration
- Deadlock and unreachable-state analysis
- Mathematical proof generation
- Reduced verification time
- Improved design reliability

## Conclusion

The project successfully demonstrates the effectiveness of formal verification using Cadence JasperGold. The FSM-based arithmetic module was verified for functionality, protocol compliance, security behavior, and deadlock-free operation. Compared to traditional simulation, formal verification provided significantly higher confidence by exploring all reachable states and uncovering hidden corner-case conditions.

## Future Enhancements

- Verification of larger SoC-level designs
- Integration with UVM-based simulation environments
- Coverage-driven formal verification
- Low-power formal verification
- Advanced security property verification